

**AreaCalculator** [index](#)  
 (version 2.0) [/mnt/c/Users/clarkes/Documents/mack/mscs/csc6301\\_software\\_design\\_doc/week2\\_from\\_python\\_to\\_java/shaun\\_clarke\\_csc6301/AreaCalculator.py](/mnt/c/Users/clarkes/Documents/mack/mscs/csc6301_software_design_doc/week2_from_python_to_java/shaun_clarke_csc6301/AreaCalculator.py)

Area calculator module for geometric area calculations.

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Provides the [AreaCalculator](#) class with utility methods for computing the area of basic geometric shapes. This version supports rectangle and triangle area calculations and includes an interactive prompt that asks the user for their name, the shape to calculate, and the required dimensions.

Typical usage would be:

```
from area_calculator import AreaCalculator
```

```
calc: AreaCalculator = AreaCalculator()
rect_area: float = calc.calculate_rectangle_area(length=10.0, width=2.0)
tri_area: float = calc.calculate_triangle_area(base=10.0, height=4.0)
```

To run interactively:

```
python area_calculator.py
```

## Classes

[builtins.object](#)  
[AreaCalculator](#)

```
class AreaCalculator(builtins.object)
    AreaCalculator() -> None
```

Provides utility methods for geometric area calculations.

An [AreaCalculator](#) instance exposes methods for computing the area of geometric shapes. The class itself holds no state; each method operates purely on the parameters provided to it. This mirrors the Java utility class pattern but in Python.

Attributes:

None. [AreaCalculator](#) is a stateless utility class.

Methods defined here:

```
__init__(self) -> None
    Create a new AreaCalculator instance.
```

The constructor takes no arguments and performs no setup, as the class is stateless. It exists to mirror the Java [AreaCalculator](#)() default constructor.

```
calculate_rectangle_area(self, length: float, width: float) -> float
    Calculate the area of a rectangle based on its length and width.
```

Multiplies the two supplied dimensions and returns the result. Both inputs are expected to be non negative floating point values representing the rectangle's sides in the same unit.

Parameters:

length : float  
 The vertical or horizontal extent of the rectangle.  
 width : float  
 The horizontal or vertical extent of the rectangle.

Returns:

float  
 The total area as the product of length and width.

```
calculate_triangle_area(self, base: float, height: float) -> float
    Calculate the area of a triangle based on its base and height.
```

Applies the standard formula (base \* height) / 2 to compute

the area of a triangle. Both inputs are expected to be non-negative floating point values measured in the same unit.

Parameters:

base : float

The length of the triangle's base.

height : float

The perpendicular height from the base to the opposite vertex.

Returns:

float

The total area as  $(\text{base} * \text{height}) / 2$ .

---

Data descriptors defined here:

\_\_dict\_\_

dictionary for instance variables

\_\_weakref\_\_

list of weak references to the object

## Functions

**main()** -> None

Run the interactive [AreaCalculator](#) prompt.

Prompts the user for their name, asks them to choose a shape rectangle or triangle, collects the required dimensions, and prints a personalized message containing the calculated area.

Notes:

Input is read with the built-in `input()` function, which is Python's equivalent of Java's `java.util.Scanner`. An `if elif` block dispatches to the correct calculation method based on the user's choice.

## Author

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